XUECHUN YU

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RESEARCH INTERESTS

GeoAI; Spatial Data Science; Multimodal Urban Sensing; Human-Centered Urban Analytics; Urban Inequality; User-Generated Geographic Information

EDUCATION

The University of Hong Kong **Hong Kong, China**
Master of Urban Design *Aug 2021 – Dec 2022*
Thesis: Analyzing Urban Spatial Structure of Retail Land Use Relative to Residential Land Use: A Case Study of Hong Kong (Grade: A–)
Thesis Advisor: Prof. Alain Chiaradia

Dalian University of Technology **Dalian, China**
B.E. in Urban and Rural Planning (five-year program) *Sep 2013 – Jun 2018*

RESEARCH EXPERIENCE

The University of Texas at Austin **Austin, TX, USA**
Research Assistant, GeoAI Research *May 2026 – Present*
- Contribute to multimodal GeoAI research integrating acoustic, visual, and geospatial data for large-scale urban environmental modeling.
- Support an ongoing research manuscript through large-scale geotagged audio processing, acoustic feature extraction, and multimodal dataset construction.
- Reproduce and evaluate audio–visual data workflows and urban environmental prediction pipelines using image embeddings and spatial variables.
- Preparing to develop and lead an independent research idea on human-centered GeoAI and geographically uneven urban experiences.

Independent Research
Neighborhood Change and Age-Specific Urban Resources *New York City | 2025 – Present*
- Construct a tract-level dataset covering approximately 2,100 census tracts to examine how gentrification reshapes access to resources supporting healthy aging.
- Apply spatial analysis, MGWR, and XGBoost-SHAP to identify spatially heterogeneous and nonlinear relationships across physical activity, cognitive stimulation, and aging-in-place resource domains.

Children's Park Affordance Actualization *Los Angeles | 2025 – Present*
- Analyze Google Maps reviews and park-level spatial data to examine how installed amenities translate into actualized affordances for children and caregivers.
- Develop UGC-based measures of amenity-use diversity and facility-level sentiment using natural language processing and spatial analysis.
- Examine representation limitations in platform-generated data, including the indirect visibility of children's experiences.

CONFERENCE PRESENTATIONS

Yu, X. (2026). “For a Younger Crowd?” Quantifying the Domain-Specific Divergence of Aging Resources in Gentrifying New York Neighborhoods. Accepted for presentation at the 2026 Association of Collegiate Schools of Planning Annual Conference.

Yu, X. (2026). From Installed to Inhabited: Measuring the Gap Between Potential and Actualized Affordances in Children’s Parks Through User-Generated Content. Accepted for presentation at the 2026 Association of Collegiate Schools of Planning Annual Conference.

SELECTED POLICY AND PLANNING RESEARCH

China Academy of Urban Planning and Design (CAUPD), Shanghai Branch **Shanghai, China**

Optimizing Urban Assessment Indicators for Child-Friendly Space Development *Dec 2024 – May 2025*

Funded by UNICEF

- Investigated the integration of child-friendly spatial criteria into China’s national urban assessment framework.
- Evaluated indicator trends and proposed age-inclusive spatial metrics for child-friendly urban governance.

National Best Practices in Child-Friendly Urban Space *Oct 2024 – Jan 2025*

Funded by the Ministry of Housing and Urban-Rural Development (MOHURD)

- Built and evaluated a national database of child-friendly spatial interventions across parks, streets, schools, and community spaces.
- Synthesized transferable design strategies and policy innovations for a national best-practice report.

Evaluation of Child-Friendly City Development in Quzhou *Aug 2024 – Feb 2025*

Funded by Quzhou Women’s Federation

- Designed a mixed-methods evaluation combining surveys, field observation, spatial mapping, indicator-based assessment, and comparative analysis.
- Produced a visual evaluation report identifying replicable practices and policy gaps.

PROFESSIONAL EXPERIENCE

China Academy of Urban Planning and Design (CAUPD), Shanghai Branch **Shanghai, China**

Urban Planner *Apr 2023 – May 2025*

- Contributed to metropolitan spatial planning, child-friendly city assessment, public-space evaluation, and policy-oriented urban research.
- Managed survey design, site assessment, stakeholder coordination, spatial analysis, and report production across government-funded projects.

Shenzhen University Institute of Urban Planning and Design **Shenzhen, China**

Assistant Planner *Jul 2018 – Jan 2020*

- Contributed to urban design, renewal, and planning projects, including the Longgang River Urban Design International Competition (2nd place).
- Conducted urban positioning, industrial analysis, and policy research for large-scale urban development projects.

TECHNICAL SKILLS

Programming & Data Analysis: Python, pandas, NumPy, Jupyter Notebook, XGBoost, SHAP

Geospatial Analysis: ArcGIS Pro, GeoPandas, spatial data processing, spatial regression, MGWR, geovisualization

Urban Data Methods: Natural language processing, user-generated content analysis, audio preprocessing, acoustic feature extraction, multimodal data processing

Design & Visualization: Adobe Illustrator, Photoshop, InDesign, AutoCAD, Rhino, SketchUp

LANGUAGES & CERTIFICATIONS

Languages: Mandarin (native); English (IELTS 7.0; TOEFL 5.0)

Certification: Junior Engineer (certified by MOHURD)

HONORS

Longgang River Urban Design International Competition, 2nd Place (2019)

Innovation and Entrepreneurship Practice Award, Dalian University of Technology (2015)